Fluids and Lubricants Specifications

All commercial MTU series (including Marine)
DDC S60 Off Highway and two-cycle engines
MTU Series 1000-1600, 1800 are not included

A001061/38E

Power. Passion. Partnership.

Table of Contents

DCL-ID: 0000005353 - 006

1 Preface

1.1 General

Used symbols and means of representation

The following instructions are highlighted in the text and must be observed:

Important information
This field contains product information which is important or useful for the user. It refers to instructions, work and activities that have to be observed to prevent damage or destruction to the material.

Note:

A note provides special instructions that must be observed when performing a task.

Fluids and lubricants

The service life, operational reliability and function of the drive systems are largely dependent on the fluids and lubricants employed. The correct selection and treatment of these fluids and lubricants are therefore extremely important. This publication specifies which fluids and lubricants are to be used.

Test standard	Designation
DIN	Federal German Standards Institute
EN	European Standards
ISO	International Standards Organization
ASTM	American Society for Testing and Materials
IP	Institute of Petroleum
DVGW	German Gas and Water Industry Association

Table 1: Test standards for fluids and lubricants

Applicability of this publication

The Fluids and Lubricants Specifications will be amended or supplemented as necessary. Prior to use, ensure that the most recent version is available. The most recent version can be called up under:

http://www.mtu-online.com/mtu/technische-info/betriebsstoffvorschriften/index.de.html

If you have further queries, please contact your MTU representative.

Warranty

Use of the approved fluids and lubricants, either under the brand name or in accordance with the specifications given in this publication, constitutes part of the warranty conditions.

The supplier of the fluids and lubricants is responsible for the worldwide standard quality of the named products.

Important information
Fluids and lubricants for drive systems may be hazardous materials. Certain regulations must be obeyed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, such as product-specific safety data sheets, statutory regulations and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for fluids and lubricants is therefore not possible within this publication.

TIM-ID: 0000018549 - 006

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. MTU accepts no responsibility whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

Preservation

All information on preservation, represervation and depreservation including the approved preservatives is available in the MTU Preservation and Represervation Specifications (publication number A001070/...). The most recent version can be called up under:

http://www.mtu-online.com/mtu/technische-info/konservierungs-und-nachkonservierungsvorschrift/index.de.html

TIM-ID: 0000018549 - 006

2 Lubricants for Four-Cycle Engines

2.1 Engine oils

Important information
Dispose of used fluids and lubricants in accordance with local regulations. Used oil must never be disposed of via the combustion engine!

Requirements of the engine oils for MTU approval

The MTU requirements for approval of engine oils for diesel engines are contained in the MTU Factory Standards MTL 5044 and MTL 5051 for first-use oils and corrosion-inhibiting oils. For two-cycle engine oils, oil approval requirements are contained in MTU Factory Standard MTL 5111. These standards can be ordered under these reference numbers.

Manufacturers of engine oils are notified in writing if their product is approved.

Approved diesel engine oils are divided into the following MTU Quality Categories:
- Oil category 1: Standard quality / Single and multigrade oils
- Oil category 2: Higher quality / Single and multigrade oils
- Oil category 2.1: Multigrade oils with a low ash-forming additive content (low SAPS oils)
- Oil category 3: Highest quality / Multigrade oils
- Oil category 3.1: Multigrade oils with a low ash-forming additive content (low SAPS oils)

Low SAPS oils are oils with a low sulfur and phosphor content and an ash-forming additive content of ≤1%.

They are only approved if the sulfur content in the fuel does not exceed 50 mg/kg. When using diesel particle filters, it is advisable to use these oils to avoid fast coating of the filter with ash particles. For exceptions, see (→ Page 10).

Selection of a suitable engine oil is based on fuel quality, projected oil drain interval and on-site climatic conditions. At present there is no international industrial standard which alone takes into account all these criteria.

Important information
The use of engine oils not approved by MTU can mean that statutory emission limits can no longer be observed. This can be a punishable offense.

Important information
Mixing different engine oils is strictly prohibited! Changing to another oil grade can be done together with an oil change. The remaining oil quantity in the engine oil system is not critical in this regard. This procedure also applies to MTU's own engine oil in the regions Europe, Middle East, Africa, America and Asia.

Important information
When changing to an engine oil in Category 3, note that the improved cleaning effect of these engine oils can result in the loosening of engine contaminants (e.g. carbon deposits). It may be necessary therefore to reduce the oil change interval and oil filter service life (one time during change).

TIM-ID: 0000018560 - 004

Special features

MTU diesel engine oils

At MTU, the following single and multigrade oils are available in the individual regions:

Manufacturer & sales region	Product name	SAE grade	Oil category	Material number
MTU Friedrichshafen Europe Middle East Africa	Diesel Engine Oil DEO SAE 15W-40	15W-40	2	20 l canister: X00070830 210 l barrel: X00070832 IBC: X00070833 Loose items: X00070835 (only on request)
	Power Guard® DEO SAE 40	40	2	20 l canister: X00062816 210 l barrel: X00062817 IBC: X00064829
MTU America Americas	Power Guard® SAE 15W-40 Off Highway Heavy Duty	15W-40	2.1	5 gallons: 800133 55 gallons: 800134 IBC: 800135
	Power Guard® SAE 40 Off Highway Heavy Duty	40	2	5 gallons: 23532941 55 gallons: 23532942
MTU Asia Asia	Diesel Engine Oil DEO SAE 15-W40	15W-40	2	18 l canister: 64247/P 200 l barrel: 65151/D
MTU Asia China	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 64242/P 205 l barrel: 65151/D
	Diesel Engine Oil - DEO 10W-40	10W-40	2	20 l canister: 60606/P
	Diesel Engine Oil - DEO 5W-30	5W-30	3	20 l canister: 60808/P
MTU Asia Indonesia	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 64242/P 205 l barrel: 65151/D
MTU India Pvt. Ltd. India	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 63333/P 205 l barrel: 65151/P
	Diesel Engine Oil - DEO 40	40	2	20 l canister: 73333/P 205 l barrel: 75151/D

Table 2:

Restrictions for applications in Series 2000 and 4000

Series 2000: Cx6, Gx6, Gx7, M41A IMO III, Mx6, M84, M94, Sx6

Series 4000: M73-M93L, N43 and N83, 4000-03 Genset (application group 3F, 3G, 3H), 4000-04 C&I, 4000-05 C&I, 4000-04 Marine, 4000-04 Rail, 4000-05 Genset, 4000-04 Oil&Gas, 4000-05 Oil&Gas

Important information
Oils in oil category 1 must not be used!

Restrictions for applications in Series 2000 M72

Important information
Mobil Delvac 1630/1640 and Power Guard® SAE 40 Off-Highway Heavy Duty must not be used!

TIM-ID: 0000018560 - 004

Restrictions on Series 4000 C, R, T application

Important information
In engines in Series 4000 C64, Cx5, T94, T94L and Tx5, only engine oils of oil category 3 or 3.1 of SAE grades 5W-40 or 10W-40 must be used! Exceptions: • For Series 4000 T, Chevron Delo 400 LE SAE 15W-40 (oil category 2.1) can also be used. • For Series 4000 T, Fleet Supreme EC SAE 15W-40 (oil category 2.1) can also be used. In engines in Series 4000 R64, R74 and R84, only engine oils of oil category 3.1 of SAE grades 5W-40 or 10W-40 must be used! The maximum oil service life is 1000 operating hours with observance of the analytical limit values for used oils!

Restrictions on Series 8000 applications

Important information
Only the following engine oils may be used: • Castrol HLX SAE 30 / SAE 40 • Exxon Mobil Delvac 1630 SAE 30 • Exxon Mobil Delvac 1640 SAE 40 • PowerGuard® SAE 40 Off-Highway Heavy Duty (material number: 5 gallons 23532941; 55 gallons 23532942) • Shell Sirius X SAE 30 / SAE 40

Restrictions for Series S60 applications

Important information
Only multigrade oils of SAE grade 15W-40 marked with index [2)] must be used. The maximum oil service life is 250 operating hours or 1 year.

Restrictions when using low SAPS oils

Important information
Oil Categories 2.1 and 3.1 may be used if the sulfur content in the fuel does not exceed 50 mg/kg. For exceptions, see (→ Page 10)

Restrictions for applications in Series 595 and 1163-01, 1163-02

Important information
Category 2 or Category 3 oils are normally stipulated for fast commercial ferries using Series 595, 1163-01 and 1163 -02 engines.

TIM-ID: 0000018560 - 004

Restrictions for applications in Series 956 TB31 / TB32 / TB33 / TB34 and 1163 TB32

Important information
Engine oils of oil categories 1, 2.1 and 3.1 are generally not approved!

Only the following engine oils are currently approved for Series 956 TB 31, TB 32, TB 33, TB 34 engines for nuclear power station applications and for Series 1163-02 TB32 engines.

Series	Oil category 2, single-grade oil	Oil category 2, multi-grade oil	Oil category 3
956 TB 31	Power Guard® SAE 40 Off-Highway Heavy Duty Mobil Delvac 1630 Mobil Delvac 1640	No approval	Shell Rimula R6 MS SAE 10W-40
956 TB 32	Power Guard® SAE 40 Off-Highway Heavy Duty Mobil Delvac 1640	No approval	Shell Rimula R6 MS SAE 10W-40
956 TB 33 Ɛ = 9	Power Guard® SAE 40 Off-Highway Heavy Duty Mobil Delvac 1640	No approval	Shell Rimula R6 MS SAE 10W-40
956 TB 33 Ɛ = 12	Power Guard® SAE 40 Off-Highway Heavy Duty Mobil Delvac 1640 Shell Sirius X 30	Lukoil Avantgarde Ultra NP 15W40	Shell Rimula R6 MS SAE 10W-40
956 TB 34	Power Guard® SAE 40 Off-Highway Heavy Duty Mobil Delvac 1640	Lukoil Avantgarde Ultra NP 15W40	Shell Rimula R6 MS SAE 10W-40
1163-02 TB 32 Emergency power, gen-set	Power Guard® SAE 40 Off-Highway Heavy Duty Mobil Delvac 1640 Shell Sirius X 30	No approval	Shell Rimula R6 MS SAE 10W-40

Table 3:

Engine oil approvals upon customer request for applications in Series 956 TB 31, TB32, TB33, TB34

The engine oil must have valid MTU approval as per MTL 5044 and a quality level of oil category 2 or 3.

For customer certification, an engine test run under the following conditions is required: Individual cylinder test run 50 hours; with positive findings the engine test run has to be carried out as follows:
- Engine runtime with specific oil: min. 50 hours (30 hours of which at min. 100% power)
- Then endoscopic examination of combustion chambers.
- Disassembly of 4 pistons (2 on engine A-side and 2 on engine B-side) for detailed results.

Use of Low SAPS oils (oil categories 2.1 and 3.1) with fuels with a maximum sulfur content of 1000 mg/kg in applications in Series 12V2000M41A, 4000M03 IMO II/III, 4000M05 IMO II/III, 8V4000M63 IMO III and 20V4000M53B IMO III

If Low SAPS oils in MTU oil categories 2.1 and 3.1 are to used in conjunction with fuels with a maximum sulfur content of 1000 ppm, in addition to the MTU approvals they must also comply with the following performance requirements:

Oil category	Specification	
	ACEA	API
2.1	E7 and E9	CJ-4 and Ci-4
3.1	E6 and E7	CI-4

Table 4:

If the Low SAPS oils meet the above-mentioned requirements, they oils are approved for a runtime up to 500 operating hours. If the maximum runtime is exceeded, the base number of the oil must be checked through regular oil analyses. If the permissible base number is undershot, the oil must be replaced.

TIM-ID: 0000018560 - 004